

Advanced algorithms, Problems number 4

Notice: Problem 2 & 6 are part of Compulsory exercise.

Problem 1

An edge coloring of a graph, is a coloring of the edges in a graph such that not two edges sharing the same endpoint has same color. The book claims that it is not too hard to show that edges in the Petersen graph can not be colored using only 3 colors. Prove/explain why.

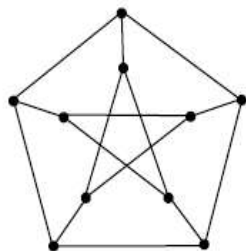


Figure 2.8: The Petersen graph. This graph is not 3-edge-colorable.

We have not seen problems like this in our lectures. You have to think!

Problem 2 (Part of compulsory)

- a) Run the dynamic programming algorithm (Chapter 3, slide 6) on the following knapsack instance with $B = 10$. Show $A(i)$ twice (before and after removing dominating pairs) for each iteration through the loop.

Item	Size	Value
1	3	5
2	2	3
3	2	2
4	4	5
5	3	4
6	1	2

- b) What is the maximum number of elements in $A(i)$ expressed in terms of B ?
- c) Let n be the number of items in the knapsack problem and let v_i be the value of item i . Furthermore let $V = \sum_{i=1}^n v_i$. What is the maximum of elements in $A(i)$ expressed in terms of V ?

Problem 3

We have the following knapsack problem where the items are ordered according to nonincreasing value.

Item	Size	Value
1	372	500 000
2	315	499 775
3	322	499 122
4	300	498 444
5	301	491 213
6	298	490 112
...	...	...
1000	99	100 000

We want a fully approximation scheme algorithm that guarantees a solution that is at least 80% of the optimal solutions.

Find ϵ and μ and show the modified values given in the table above. Hint: Chapter 3, slides 10 – 16.

Problem 4 (Scheduling)

We want to schedule jobs according to Chapter 3, Slide 17. Given 13 jobs with corresponding processing time:

j	1	2	3	4	5	6	7	8	9	10	11	12	13
P _j	20	17	15	14	12	10	9	7	7	5	3	3	1

Let $m = 3$ and $k = 4$.

- What determines if a job is long or short?
- Show the schedule after applying the algorithm on slide 19.

Problem 5 (binpacking)

- Use the First-Fit-Decreasing (FFD) algorithm to pack the following list of items: {0.60, 0.51, 0.50, 0.35, 0.33, 0.33, 0.30, 0.30, 0.15, 0.10, 0.05}.
- Argue that the solution found by FFD is optimal in the instance above.
- Given the instance (we have just shown just the beginning) $I = \{0.68, 0.67, 0.66, 0.65, 0.63, 0.63, 0.60, 0.60, 0.59, 0.58, 0.57, 0.55, 0.48, 0.47, 0.45, \dots\}$. Show the start of the corresponding rounded instance I' obtained by linear grouping. Use $k = 3$ (this is very small for practical problems since we have $k = \lfloor \epsilon \text{SIZE}(I) \rfloor$).

Problem 6 & 7 are from Chapter 4

Problem 6 (Part of compulsory)

Consider the problem of minimizing the sum of completion times on a single machine given the data below (start of Chapter 4):

Job number	Release time	Processing time
1	0	6
2	0	5
3	1	3
4	2	1
5	3	4
6	4	2
7	19	2

Use the approximation algorithm based on the Shortest Remaining Processing Time for preemptive scheduling (Chapter 4, slide 7) to find an approximate solution to the nonpreemptive problem.

Problem 7 (binpacking)

- We have 120 elements of size 0.34 and 200 elements of size 0.21. Enumerate possible configurations (see slides 16 – 19, Chapter 4).
- Formulate the instance above as an integer program.